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highly worthy of engaging the attention of the young gentleman. Dancing, fencing, running, horteman- flip, the management of the mofics, and the motions of military glimphes, are more at them worthy of occupying his time, at proper feafons. It is unenceffary to point out the advantages which he may be view from claiming it their own in to be pretty generally understood. Perhaps our men of forture would be afhamed to make use of their legs for running; but occasions may occur, on which even this humble do compliment may be useful. Though we with not to fee the young man of fortune become a jockey yet to be able to make a greatful appearance on both back, and to manage his boric with dexterity, will not be between of his flation and character. If true of public danger though arise, and the fate founds call for the fervies of her objects against any hostile attack; they whole rank and fortune place them in the most eminent fiations will be felt expected to hand forth; but if unacquated with thofe exercises which are connected with the military art, what a pitiful figure mult they make in the camp, or on the field of battle?

Lav.

Natural philosophy, as being both highly ufeful and entertaining, as well worthy of the attention of all who can talked to appropriate anayment of their time to fictific purifuits; to the man of fortune, a taille for mutual philopholy might often procure the mind de-lightful environment. To trace the wonder of a planetary tytions, to mark the proceeds of vegetation, to examine all the properties of that the elements which we bearbe, to trace the laws by which all the different elements are confined to their proper funcions, and above all to apply the principles of natural philopholy in the cultivated of the ground, and amendments which might agreeably and innocently occupy many of the hitline hours of the man who enjoys a friend and independent fortune.

Philophy.

7. Milbury and morals

Neither do we fuppole civil history and the pria-ciples of animals to be overlooked. Without being anaquished with theftsy, how could not yunk or accurate knowledge of the laws and political contributions of his country be acquired by the young generation! Hilory expects to war observation the fortune and the echo of other human beings, and thus fuppies in force mea: fine the place of experience; it teacher graduate, and affords exercise to the moraliente. When history condelends to take notice of individuals, they see all making always flesh a have been embent for virtue, for abilities, or for the rank which, they held in life; to the rich and great it ought to freak with peculiar effficiency, and they ought to be carefully invited to listen to its voice.

Such then is the manner in which we with the education of young men of mind, and forture to be conducts; in order that they may be prepared for en-joying their opulence and honours with becoming digintegrity. Let them be early inured to habit of vigorous

indufry and perfevering firmness, by pulling through a Education. regular course of claffical learning in a free (chool; let them play and converser with their equals, and not be permitted to form high ideas of their

own importance, nor to dominate over ferrets or infortors; Let them be carefully instructed to the principles of monality and religion. Let now be taught the mainly exercices: Let them be carefully informed of the nature of the political constitution of their country, and of the ex-tent of their spirit and politick rights which is focuses to them and their fellow-officers; Let them be called to trace the annals of mandat through the records of history; to mark the appearances and operations of nature, and to amute themselves by purifying thefe to their general earlier. We lay nothing of causing the young man of fortune to learn some mechanical art; We think fell in a mechanical arm might now and then afford him an innovacant and plading amefiens; but we do one consider it is absolutely necessary, and therefore do soe infinit on his acquiring it. With those accomplishments we hope he might become an useful member of Society, might adorn the rank and fortune to which he is born, and might find wealth than high flation a building, not a curic. It is peculiarly unfair toame for our age and country, that people of rank and forture are not to illudious that their children ac-quire there as the more superficial accomplillments.

VII. In the Education of People Affordford for a Mercan- tile deployments and join the number Occupations in Life is particularly controlled with Literatures.

Wars modern literature in a lefs flourishing flare; were the English and French language along with fewer tenents social, albidental, and philosophical compositions; we might perhaps infinit on it as ne-efficacy to give the boy, who is designed for a mercun-tile emplements, a classification function. At present this does not appear absolutely necellary; yet we do not perform to forbid it as improperly. Even the Elegantile members will never lead reason to repon, his he cannot, vying been introduced to the acquaintance of Plato and Cicero. But fall, of the circumnation of the parents, or any other part radios, should render it inconvenient to fend the young man who is intand- ed to trade in a few feebed to fluly the ancient languages, means may be really adopted to make up for his lofa. Confine him not to writing and accounts alone. Thief, though, appropriately useful to the mer-clans, have no great power to refrain the force of evil pallions, or to injure the mind with generous and written feminements. Though you burger him not with Latin and Greek, yet drive to injure him with a tale for deaf knowledge and for elegant literature. Some of the pureft and molt elegant of our posts, the excellent periodicals works which have appeared in new languages, facts as the Spectator, the Adventures, the Mirror, and the composition of our fresh blibrarians, together with tone of the belt translations of the claf-fas which are profits, thats you may with great pro- pristry put into go-b hands. They will teach him how to think and realon jolly, and to express himself in conversation or in writing with coretherla and elegance, they shall online and publish his mind, and raife him above low and greats pleafures. And as soon, who has any occasion to freak or write, ought to be entirely ignorant of the principles of grammar, you will Egypt. over the continent, they had become a great and powerful people and from their opposite dispositions and manners, became very frequently enemies to the Egyptians. "To me Salaris our author whether the definition of Thebes in Upper Egypt, is most celebrated by Horror for its magnificence. But this certainly cannot be the epic; for Honor were long at terre time of Jodphs, and we find that even then the Egyptians had the fhepherds in abhorrence, in all probability because they had been grietically oprified by them. Mr Bruce counts are insured if the people; the filt, that of Salats already mentioned, who overtheve the first dynasty of Egyptian bags from Mexico, and displayed Thebes: the second was that of Sabasco or So 2 for according to him this was not the name as single prices, but of a people against _fas, Ephohele and_ the third, after the building of Memphis, where 240,000 of them were befeged as above mentioned. But accounts in the blind are recently identified in the highest degree? He here is it possible that the _third_ invasion, antecedent to the building of Jerusalem, could be policerion to the _feared_, if the latter happened only in the way of Hezschal?

In their early ages, however, it would feem that the kingdom of Egypt had been very powerful and its do anunion very well extraction, since we had it fady, that the _Batiriana_ resulted from Olymandysan, another Egyptians king of every high authority, and of whole wealth, the much moreulous accounts are given.

After an unknown interval of time from this monarchy existed Sofifris. He on the thrill great war-arian whole conquests are recorded with any other words of difficultés. In what age of the world he lived, is uncertain. Some choralogers, among those is Sir!a Newtown, are still opinion, thus he is the Sefer Shilhak, who took Jerulidam in the origin of Re-holone the fun of Solomon. Others, however, phen him much earlier) and Mr Whition will have him to be the Pharash who refuted to part with the IF realches, and was at fall drowned in the K Sea. Mr Bryant endeavors to pose that no lich perform ever exfcited; but that in his history, as well that of many ancient know, we have an advantage of that of the Cultivite or Ablybunians, who spread themselves over great part of the known word, and every where brought the people in disposition

to them. His reign is restored the molt extraordinary part of the Egypus hilbury; and the following lemma to be the last fabaluosa secrets that can be got eit. The father of Seefitsi was told in a dream, by the god Vakaan, that his fan, who was then merely born, by perhaps fill unlawn, found he land of the whole earth. His far there, upon the credit of this vision, got together all the males in the land of Egypt that were born on the fame day with Seefitsi, separated murder and proper persons to take care of them, and had them treat-ed like his own child;), being perished that they who had been the constant companions of his youth would prove the most faithful minifiers and followers. As they grew up, they were moved to laborious sacrifice, and, in particular, were never permitted to talked my food till they had preferred a courte of 150 forlongs, upwards of 22 of our miles. When the old king imagined they were sufficiently educated in the martial way he deliged them to follow, they were sent by way trial of their abilities against the Ambana. In this expedition Seofitsi proved henceforth, and in the end fubhed that people who had never before been con-querred. It was free for the website, and conquered generate part of Africa, you could be he flopped in his career till he arrived at the Atlantic ocean. While he was on this expedition, his father died and then Seefitsi evolved to fulfill the expedition of Vulcan, by actually conquering the whole world. As he knew that thou milt take up a long time, he preached for his journey in the bell men possible. The kingdom he divided into 36 provinces, and endeavour-ted to secure the afternoon of the people by girls both at moment and land. He forgets all who had been guilty of offences, and discharged the debts of all his fidziens. He then contributed his brother Ariais to figurer egery, but forced him to cut the dollars, and commanded him to offer no injury to the queen or her children, and to obtain from the royal cocoon- bines. This may conflict of 600,000 horie, 240,000 horse, and 27,000 charriots. Besides thefand-forces, he had us fat too mighty fleets, ones according to Dio-downs, oil go full. Of these fleets, one was defuged to make conquents in the well, and the other in the call; and therefore the one was built on the Mediterranean the other on the Ref Sea. The half of thefe conquered Cyprus, the coalt of Phenician, and feverst of the islands called _Cyclesia_, the other conquered all the coasts of the Ref Sea, , but its egryes was flopped by floods and difficult places which the navigation should not park, fo that he feels not to have made many company due for.

With the land-forces Seofitsi marched against the Ethiopian and Trojoldietten, whom are overcame, and obliged him to pay him a tritir of gold, ohony and ivory. From theme he proceeded as far as the pro-mony of Uma, which lay near the finest of Babelmaded, when lay the pillar with an inforticipation in faced characters. He then marched on to the country where cinnamon grows, or as leath to form country where cinnamon is that time to brought, probably fome place in Indias and here, he in like manner fet y pilnas, what were to be free for many ages after. At to his farther conquels, it is agreed by almost all authors of antiquity, that he over-one and pillaged the whole continent of Arab, and fome part of Europe. He crotified the Ganges, and credited pillars on its banks; and from theme he is fail to now reached outward to the very extremist yet the African continent. Returning from friends, he invaded the Sey- thana and Thincians; but all authors do not give that he conquered them. Some some affirm, that he was overthrown by them with great daughters, and ob-glied to abandon a great part of his body and military forces. But whether he had good or bad faces in thefe parts, it in a common opinion that he fetted a colony in Colombia. Freedomey, who gives the most popular account of the conquests of this monarch, does not fay whether the colony was designated, planted by Seefitsi's, or whether part of his army lowered behind the rell, and took up their residence in that region. From his own knowledge, he afferts, that the inhabitants of that country were undoubtedly of Egyptian defects. This was evident from the personal referendure they bore to the Egypu Effaying weighed, and made of paper not impregnated with alum. It is to be dried at frift with a gentle heat, but afterwards expelled for five minutes to a beat of 10:00 On cooling, it to the weights together with the fiter; the brown weight of which mult afterwards be fobricked. The fedmena is both included in bottle for a filter than once impregnated with false matter cannot be freed from it again without great difficulty, especially if an internal of one burns intervents.

The alkali mute ufc of in Mr Bergantis experiments, was that of fols faturated with aerial acid, also ploughedicated skills needed by defibrating equal weights of poor mice and create on tartrationally mixed together; the reifulaum is the common white flux. Half an onces of this is indifferent in half a quadrans of distilled water. This is the adda, as a digital heat, two ounces of Prulfian blue, carefully avoiding (ch) an evervidence upon throw any thing or, which cellily happen if the quantity be too large. The pig-ment footo lodes its beautiful blue colour, growing not red but blacks, which shows that a decomposition has taken place. The Prussian blue rule of in his expert mentioned in 100 parts only 33 of the pigment and 57 of the earth is low. If we employ the blue male exact way alum, 22 grams of K will saturate the half onces of

alkaline fat more completely than the two causes of the final already described. But in whatever manner the operation is performed, after the addition of the hall quantity, the whole mult to ex-paid to a stronger diguling heat, and differed with wooden faults. If the liquor be too much diminishely on proportion, the deficit mult be figured by adding more water. When the liquor becomes clear the réiduum mult be collected upon filtering paper, and gradually washed with warm water until all the fullable part is intended, where, if the proportion has been properly conducted, the filtered liquor amounts to a whole quamings, of a brown yellow color, and fo well futurated with colouring matters, that it does not change the colour of paper tinged with Brafl wood. This historic, however, contains a funqu-tivity of Prulfian blue, about 3d, No a ctwy of the al-kali. Thefe should be previously farparted by an acid, or, which is better, corrected by flutstanding from the weight of the delinattor of clay grounds for each quasrans of the likivium. When we with to examine the colour of the products exactly, however, the likivium we employ mult mustically be well departedly, for by neglecting this precaution we may easily perturbeour-fleew that any mist precipitated by the likivion has a blue color. When we out with to ascertain the weight, the likivium, having the final proportion of Prulfian blue becomes, may be greatest, but fill the proper correction mult ultimately do made ufc of; for the precipitating said is wont to impair the quali-tion of the likivium, and even to deliver them altogether, especially in a warm temperature. Calcorous cards, whether in its mild or canific fixate, is also capable of ablifting a coloured fuhlance from iron and other metals.

Int the precipitation of metal the metal, it is to be ordered, that the card of the foliation ought to be somewhat predominant; but any confidable excess mult be corrected occasionally, either by alkali, water, or spirit of wine.

In the following experiments an effay ewt. was always employed, unlels where it is expressly mentioned otherwise; conclusions sufficiently accurate may in- dred be obtained from 37th, my functions from used in Mr smaller quantities. In their cafes our author mentions Respond a the initial quantity, applying to them thoir formula of reperi- calculation which are founded on the mutual purpose results. tons of the proximate principles confluiting metallic fails. By us early fuhlition, one time broke may be useful by those who employ 5 or 4 vet. We now come to defer the method of elifying the ones of the partities mental.

1. _Ores of Gold._ This metal occurs in the bowels of the earth native, poffelling a complete metallic forms of the area although in general the small particles of it are for in-of gold.

Forperfected various mattons that they are entirely invisible. It is also found mineralized, or united with fulpharby means of iron or from other me- inclined. Therefore ten types of cost we feel counterly fe-partably.

Native gold is very felddom, if every, free from hote- Native rogreous matters; the molt uful mixtures are cop-gold, per, silver, and formtimes iron. The first of the remaining the medium, and may be particularly calefed by diffusing the gold in auguegies, and precipitating by by martial virtvel; the second falls during the following yielding a flatelry, which, being washed and dried, allows the weight of the fiber cou-tained, and the iron may be discovered by pholgills- coated alkali. The precipitate occurred by martial vis-fold in pure gold in its metallic base, but very definitely divided, and therefore its weight requires so correction.

Hence it appears how small a portion of gold in herrent in the ones of other metals may be extracted; besides, it follows containing the molt minute parcle of gold initially produces the purple precipitate of Caffius, with a foliation of its property pre-parallel.

As to the ore which contains gold adhering to and Ores of this furrounded by flony particles, r. We mult reduce a deter-metal pro-mined weight to an impulsive powdery texture and pretty da-alitarian. Then let the powdery weighted forvolume, for boiled in aqua regis, as long as any thing is taken, by the mediums), after which, let the exhausted ore, well washed, be collected, refecuted to spinism, and weighed. Let the clear features (the colour of which, in forens above affords a method of judging) be precipitated in the natural way by martial virtvel; the precipitate well washed, dried, and weighed, flows the gold, which, added to the weight of the exhausted ore, might to be equal to the original weight, unless fenome what has been dispersed by the pulverification, or beliefs force of the mattons has entered the meltdown. The former of thefts delivered by comparing the weights before and after pulverification; the latter by precipitants.

When grains of gold are mixed with loofe earthy particles, they are sometimes easily ferated by mec-channel application of water.

When the metal is mineralised by fulphur, as in the golden pyster, let too an more effay ctw, reduced to arilized by powder be gently boiled in the nitrated acid, or rather fulphur, digested in a heat of 5080ř, left the fulphur should be delayed. It is even neceffary to employ a more gentle brat for this purports, that the fulphurcous pr-ticles

Source record

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Education. highly worthy of engaging the attention of the young gentleman. Dancing, fencing, running, horſemanſhip, the management of the muſket, and the motions of military diſcipline, are none of them unworthy of occupying his time, at proper ſeaſons. It is unneceſſary to point out the advantages which he may derive from dancing; theſe ſeem to be pretty generally underſtood. Perhaps our men of fortune would be aſhamed to make uſe of their legs for running; but occaſions may occur, on which even this humble accompliſhment may be uſeful. Though we wiſh not to ſee the young man of fortune become a jockey; yet to be able to make a graceful appearance on horſeback, and to manage his horſe with dexterity, will not be unworthy of his ſtation and character. If times of public danger ſhould ariſe, and the ſtate ſhould call for the ſervices of her ſubjects againſt any hoſtile attack, they whoſe rank and fortune place them in the moſt eminent ſtations will be firſt expected to ſtand forth; but if unacquainted with thoſe exerciſes which are connected with the military art, what a pitiful figure muſt they make in the camp, or on the field of battle?

71 Law. As the man of fortune may perhaps enjoy by hereditary right, or may be called by the voice of his fellow-citizens, to a ſeat among the legiſlative body of his country; he ought in his youth to be carefully inſtructed in the principles of her political conſtitution, and of thoſe laws by which his own rights and the rights of his fellow-citizens are determined and ſecured.

72 Philoſophy. Natural philoſophy, as being both highly uſeful and entertaining, is well worthy of the attention of all who can afford to appropriate any part of their time to ſcientific purſuits; to the man of fortune, a taſte for natural philoſophy might often procure the moſt delightful entertainment. To trace the wonders of the planetary ſyſtems, to mark the proceſs of vegetation, to examine all the properties of that fine element which we breathe, to trace the laws by which all the different elements are confined to their proper functions, and above all to apply the principles of natural philoſophy in the cultivation of the ground, are amuſements which might agreeably and innocently occupy many of the leiſure hours of the man who enjoys a ſplendid and independent fortune.

73 Hiſtory and morals. Neither do we ſuppoſe civil hiſtory and the principles of morals to be overlooked. Without being acquainted with theſe, how could any juſt or accurate knowledge of the laws and political conſtitution of his country be acquired by the young gentleman? Hiſtory expoſes to our obſervation the fortune and the actions of other human beings, and thus ſupplies in ſome meaſure the place of experience; it teaches prudence, and affords exerciſe to the moral ſenſe. When hiſtory condeſcends to take notice of individuals, they are almoſt always ſuch as have been eminent for virtue, for abilities, or for the rank which they held in life; to the rich and great it ought to ſpeak with peculiar efficacy, and they ought to be carefully invited to liſten to its voice.

Such then is the manner in which we wiſh the education of young men of rank and fortune to be conducted, in order that they may be prepared for enjoying their opulence and honours with becoming dignity. Let them be early inured to habits of vigorous induſtry and perſevering firmneſs, by paſſing through a regular courſe of claſſical learning in a free ſchool; let them play and converſe with their equals, and not be permitted to form high ideas of their own importance, nor to domineer over ſervants or inferiors: Let them be carefully inſtructed in the principles of morality and religion: Let them be taught the manly exerciſes: Let them be carefully informed of the nature of the political conſtitution of their country, and of the extent of thoſe civil and political rights which it ſecures to them and their fellow-citizens: Let them be called to trace the annals of mankind through the records of hiſtory; to mark the appearances and operations of nature, and to amuſe themſelves by purſuing theſe to their general cauſes. We ſay nothing of cauſing the young man of fortune to learn ſome mechanical art: We think ſkill in a mechanical art might now and then afford him an innocent and pleaſing amuſement; but we do not conſider it as abſolutely neceſſary, and therefore do not inſiſt on his acquiring it. With thoſe accompliſhments we hope he might become an uſeful member of ſociety, might adorn the rank and fortune to which he is born, and might find wealth and high ſtation a bleſſing, not a curſe. It is peculiarly unfortunate for our age and country, that people of rank and fortune are not ſo ſtudious that their children acquire theſe as the more ſuperficial accompliſhments. Education.

VII. *On the Education of People deſigned for a Mercantile Employment, and for the humbler Occupations in Life not particularly connected with Literature.*

Were modern literature in a leſs flouriſhing ſtate; were the Engliſh and French languages adorned with fewer eminent poetical, hiſtorical, and philoſophical compoſitions; we might perhaps inſiſt on it as neceſſary to give the boy, who is deſigned for a mercantile employment, a claſſical education. At preſent this does not appear abſolutely neceſſary; yet we do not preſume to forbid it as improper. Even the merchant will ſcarce find reaſon to repent his having been introduced to the acquaintance of Plato and Cicero. But ſtill, if the circumſtances of the parent, or any other juſt reaſon, ſhould render it inconvenient to ſend the young man who is intended for trade to a free ſchool to ſtudy the ancient languages, means may be eaſily adopted to make up for his loſs. Confine him not to writing and accounts alone. Theſe, though particularly uſeful to the merchant, have no great power to reſtrain the force of evil paſſions, or to inſpire the mind with generous and virtuous ſentiments. Though you burden him not with Latin and Greek, yet ſtrive to inſpire him with a taſte for uſeful knowledge and for elegant literature. Some of the pureſt and moſt elegant of our poets, the excellent periodical works which have appeared in our language, ſuch as the Spectator, the Adventurer, the Mirror, and the compoſitions of our Britiſh hiſtorians, together with ſome of the beſt tranſlations of the claſſics which we poſſeſs; theſe you may with great propriety put into his hands. They will teach him how to think and reaſon juſtly, and to expreſs himſelf in converſation or in writing with correctneſs and elegance: they will refine and poliſh his mind, and raiſe him above low and groſs pleaſures. And as no man, who has any occaſion to ſpeak or write, ought to be entirely ignorant of the principles of grammar, you will 74 Elegant literature.

Figure 1: *
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Egypt. over the continent, they had become a great and powerful people; and from their oppoſite diſpoſitions and manners, became very frequently enemies to the Egyptians. To one Salatis our author aſcribes the deſtruction of Thebes in Upper Egypt, ſo much celebrated by Homer for its magnificence. But this certainly cannot be the caſe; for Homer wrote long after the time of Joſeph: and we find that even then the Egyptians had the ſhepherds in abhorrence, in all probability becauſe they had been grievouſly oppreſſed by them. Mr Bruce counts three invaſions of theſe people; the firſt, that of Salatis already mentioned, who overthrew the firſt dynaſty of Egyptian kings from Menes, and deſtroyed Thebes: the ſecond was that of Sabacco or *So*; for according to him this was not the name of a ſingle prince, but of a people, and ſignifies *ſhepherds*: and the third, after the building of Memphis, where 240,000 of them were beſieged as above mentioned. But accounts of this kind are evidently inconſiſtent in the higheſt degree; for how is it poſſible that the *third* invaſion, antecedent to the building of Jeruſalem, could be poſterior to the *ſecond*, if the latter happened only in the days of Hezekiah?

In theſe early ages, however, it would ſeem that the kingdom of Egypt had been very powerful and its dominion very widely extended; ſince we find it ſaid, that the *Bactrians* revolted from Oſymandyas, another Egyptian king of very high antiquity, and of whoſe wealth the moſt marvellous accounts are given.

After an unknown interval of time from this monarch, reigned Seſoſtris. He was the firſt great warrior whoſe conqueſts are recorded with any degree of diſtinctneſs. In what age of the world he lived, is uncertain. Some chronologers, among whom is Sir Iſaac Newton, are of opinion, that he is the Seſac or Shiſhak, who took Jeruſalem in the reign of Rehoboam the ſon of Solomon. Others, however, place him much earlier; and Mr Whiſton will have him to be the Pharaoh who refuſed to part with the Iſraelites, and was at laſt drowned in the Red Sea. Mr Bryant endeavours to prove that no ſuch perſon ever exiſted; but that in his hiſtory, as well as that of many ancient heroes, we have an abridgement of that of the Cuſhites or Babylonians, who ſpread themſelves over great part of the known world, and every where brought the people in ſubjection to them. His reign is reckoned the moſt extraordinary part of the Egyptian hiſtory; and the following ſeems to be the leaſt fabulous account that can be got of it. The father of Seſoſtris was told in a dream, by the god Vulcan, that his ſon, who was then newly born, or perhaps ſtill unborn, ſhould be lord of the whole earth. His father, upon the credit of this viſion, got together all the males in the land of Egypt that were born on the ſame day with Seſoſtris; appointed nurſes and proper perſons to take care of them, and had them treated like his own child; being perſuaded that they who had been the conſtant companions of his youth would prove the moſt faithful miniſters and ſoldiers. As they grew up, they were inured to laborious exerciſes; and, in particular, were never permitted to taſte any food till they had performed a courſe of 180 furlongs, upwards of 22 of our miles. When the old king imagined they were ſufficiently educated in the martial way he deſigned them to follow, they were ſent by way of trial of their abilities againſt the Arabians. In this Egypt. expedition Seſoſtris proved ſucceſsful, and in the end ſubdued that people who had never before been conquered. He was ſent to the weſtward, and conquered the greateſt part of Africa; nor could he be ſtopped in his career till he arrived at the Atlantic ocean. Whilſt he was on this expedition, his father died; and then Seſoſtris reſolved to fulfil the prediction of Vulcan, by actually conquering the whole world. As he knew that this muſt take up a long time, he prepared for his journey in the beſt manner poſſible. The kingdom he divided into 36 provinces, and endeavoured to ſecure the affections of the people by gifts both of money and land. He forgave all who had been guilty of offences, and diſcharged the debts of all his ſoldiers. He then conſtituted his brother Arais the ſupreme regent; but forbad him to uſe the diadem, and commanded him to offer no injury to the queen or her children, and to abſtain from the royal concubines. His army conſiſted of 600,000 foot, 24,000 horſe, and 27,000 chariots. Beſides theſe land-forces, he had at ſea two mighty fleets; one, according to Diodorus, of 400 ſail. Of theſe fleets, one was deſigned to make conqueſts in the weſt, and the other in the eaſt; and therefore the one was built on the Mediterranean and the other on the Red Sea. The firſt of theſe conquered Cyprus, the coaſt of Phœnicia, and ſeveral of the iſlands called *Cyclades*: the other conquered all the coaſts of the Red Sea; but its progreſs was ſtopped by ſhoals and difficult places which the navigators could not paſs, ſo that he ſeems not to have made many conqueſts by ſea.

With the land-forces Seſoſtris marched againſt the Ethiopians and Troglodites; whom he overcame, and obliged them to pay him a tribute of gold, ebony, and ivory. From thence he proceeded as far as the promontory of Dira, which lay near the ſtraits of Babelmandel, where he ſet up a pillar with an inſcription in ſacred characters. He then marched on to the country where cinnamon grows, or at leaſt to ſome country where cinnamon at that time was brought, probably ſome place in India; and here he in like manner ſet up pillars, which were to be ſeen for many ages after. As to his farther conqueſts, it is agreed by almoſt all authors of antiquity, that he over-ran and pillaged the whole continent of Aſia, and ſome part of Europe. He croſſed the Ganges, and erected pillars on its banks; and from thence he is ſaid to have marched eaſtward to the very extremity of the Aſiatic continent. Returning from thence, he invaded the Scythians and Thracians; but all authors do not agree that he conquered them. Some even affirm, that he was overthrown by them with great ſlaughter, and obliged to abandon a great part of his booty and military ſtores. But whether he had good or bad ſucceſs in theſe parts, it is a common opinion that he ſettled a colony in Colchis. Herodotus, however, who gives the moſt particular account of the conqueſts of this monarch, does not ſay whether the colony was deſignedly planted by Seſoſtris; or whether part of his army loitered behind the reſt, and took up their reſidence in that region. From his own knowledge, he aſſerts, that the inhabitants of that country were undoubtedly of Egyptian deſcent. This was evident from the perſonal reſemblance they bore to the Egyp-

Figure 2: *
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Effaying. weighed, and made of paper not impregnated with alum. It is to be dried at firſt with a gentle heat, but afterwards expoſed for five minutes to a heat of 100°. On cooling, it is to be weighed together with the filter; the known weight of which muſt afterwards be ſubtracted. The ſediment is beſt waſhed in a bottle; for a filter when once impregnated with ſaline matter cannot be freed from it again without great difficulty, eſpecially if an interval of ſome hours intervenes.

7 Alkalies nitre and phlogiſticated, uſed by Mr Bergman.

The alkali made uſe of in Mr Bergman's experiments, was that of ſoda ſaturated with aerial acid. His phlogiſticated alkali is made by deflagrating equal weights of pure nitre and cream of tartar intimately mixed together; the reſiduum is the common white flux. Half an ounce of this is diſſolved in half a quadrans of diſtilled water. To this he adds, in a digeſting heat, two ounces of Pruſſian blue, carefully avoiding ſuch an effervеſcence as may throw any thing over, which eaſily happens if the quantity be too large. The pigment ſoon loſes its beautiful blue colour, growing not red but black; which ſhows that a decompoſition has taken place. The Pruſſian blue uſed in his experiments contained in 100 parts only 23 of the pigment and 77 of the clay; ſo that if we employ the blue made without any alum, 221 grains of it will ſaturate the half ounce of alkaline ſalt more completely than the two ounces of the kind already deſcribed. But in whatever manner the operation is performed, after the addition of the laſt quantity, the whole muſt be expoſed to a ſtronger digeſting heat, and ſtirred with a wooden ſpatula. If the liquor be too much diminiſhed by evaporation, the defect muſt be ſupplied by adding more water. When the liquor becomes clear, the reſiduum muſt be collected upon filtering paper, and gradually waſhed with warm water until all the ſoluble part is extracted; when, if the operation has been properly conducted, the filtered liquor amounts to a whole quadrans, of a browniſh yellow colour, and ſo well ſaturated with colouring matter, that it does not change the colour of paper tinged with Braſil wood. This lixivium, however, contains a ſmall quantity of Pruſſian blue, about 4 lb. to a cwt. of the alkali. Theſe ſhould be previouſly ſeparated by an acid, or, which is better, corrected by ſubtracting from the weight of the ſediment 16 eſſay pounds for each quadrans of the lixivium. When we wiſh to examine the colour of the precipitate exactly, however, the lixivium we employ muſt neceſſarily be well depurated; for by neglecting this precaution we may eaſily perſuade ourſelves that any metal precipitated by the lixivium has a blue colour. When we only wiſh to aſcertain the weight, the lixivium, having the ſmall proportion of Pruſſian blue intermixed, may be employed: but ſtill the proper correction muſt ultimately be made uſe of; for the precipitating acid is wont to impair the qualities of the lixivium, and even to deſtroy them altogether, eſpecially in a warm temperature. Calcareous earth, whether in its mild or cauſtic ſtate, is alſo capable of abſtracting a coloured ſubſtance from iron and other metals.

8 How to precipitate metals by metals.

In the precipitation of metals by metals, it is to be obſerved, that the acid of the ſolution ought to be ſomewhat predominant*; but any conſiderable exceſs muſt be corrected occaſionally, either by alkali, water, or ſpirit of wine.

In the following experiments an eſſay cwt. was always employed, unleſs where it is expreſsly mentioned otherwiſe: concluſions ſufficiently accurate may indeed be obtained from 25 lb. nay ſometimes from ſmaller quantities. In theſe caſes our author mentions the uſual quantity; applying to them thoſe formulæ of calculation which are founded on the mutual proportions of the proximate principles conſtituting metallic ſalts. By an eaſy ſubſtitution, the ſame formulæ may be uſed by thoſe who employ ½ or ¼ cwt. We now come to deſcribe the method of eſſaying the ores of the particular metals. Effaying.

9 Weights uſed in Mr Bergman's experiments.

10 How to eſſay the ores of gold.

1. *Ores of Gold.* This metal occurs in the bowels of the earth native, poſſeſſing a complete metallic form, although in general the ſmall particles of it are ſo interſperſed in various matrices, that they are entirely inviſible. It is alſo found mineralized, or united with ſulphur by means of iron or ſome other metal. Theſe two ſpecies of ore we ſhall conſider ſeparately.

11 Native gold.

Native gold is very ſeldom, if ever, free from heterogeneous matters; the moſt uſual mixtures are copper, ſilver, and ſometimes iron. The firſt of theſe remains in the menſtruum, and may be ſeparately collected by diſſolving the gold in aqua-regia, and precipitating it by martial vitriol: the ſecond falls during the ſolution, yielding a ſalited ſilver; which, being waſhed and dried, ſhows the weight of the ſilver contained; and the iron may be diſcovered by phlogiſticated alkali. The precipitate occaſioned by martial vitriol is pure gold in its metallic ſtate, but very ſubtilely divided, and therefore its weight requires no correction.

Hence it appears how ſmall a portion of gold inherent in the ores of other metals may be extracted; beſides, a ſolution containing the moſt minute particle of gold inſtantly produces the purple precipitate of Caſſius, with a ſolution of tin properly prepared.

12 Ores of this metal properly ſo called.

As to the ore which contains gold adhering to and ſurrounded by ſtony particles, 1. We muſt reduce a determined weight to an impalpable powder by triture and elutriation. Then let the powder, weighed a ſecond time, be boiled in aqua regia, as long as any thing is taken up by the menſtruum; after which, let the exhauſted ore, well waſhed, be collected, exſiccated to ignition, and weighed. Let the clear ſolution (the colour of which, in ſome degree, affords a method of judging) be precipitated in the uſual way by martial vitriol; the precipitate well waſhed, dried, and weighed, ſhows the gold, which, added to the weight of the exhauſted ore, ought to be equal to the original weight, unleſs ſomewhat has been diſperſed by the pulveriſation, or unleſs ſome of the matrice has entered the menſtruum. The former of theſe is diſcovered by comparing the weights before and after pulveriſation; the latter by precipitants.

When grains of gold are mixed with looſe earthy particles, they are ſometimes eaſily ſeparated by mechanical application of water.

13 Gold mineralized by ſulphur.

When the metal is mineraliſed by ſulphur, as in the golden pyrites, let one or more eſſay cwts. reduced to powder be gently boiled in the nitrous acid, or rather digeſted in a heat of 50°—80°, leſt the ſulphur ſhould be deſtroyed. It is even neceſſary to employ a more gentle heat for this purpoſe, that the ſulphureous particles,

Figure 3: *
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